SAMARTH

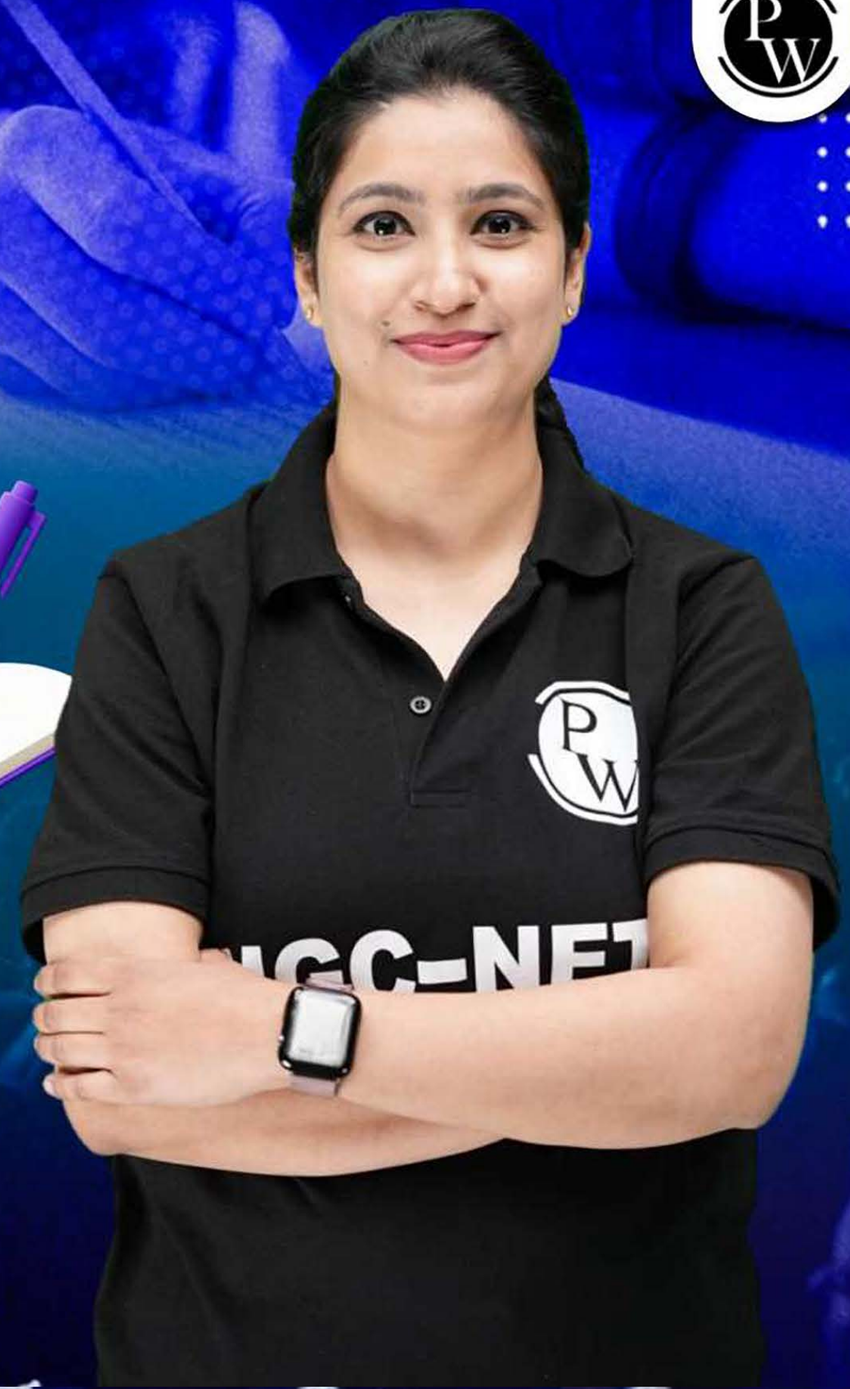
CUET PG 2027

Lecture - 07 | Computer Science

Unit 03 : Digital Fundamental

Topic : Digital Computers

By- Puneet Kaur



Topics *to be covered*

Digital Computers



Topic : Basic Computer Fundamentals and Organization

- **Components of Computer**: Understanding the various components of a computer is essential for grasping how these machines function. Computers consist of hardware and software elements, each playing a crucial role in their operation. Check complete details of Components of Computer here.

Topic : Basic Computer Fundamentals and Organization

Components of Computer

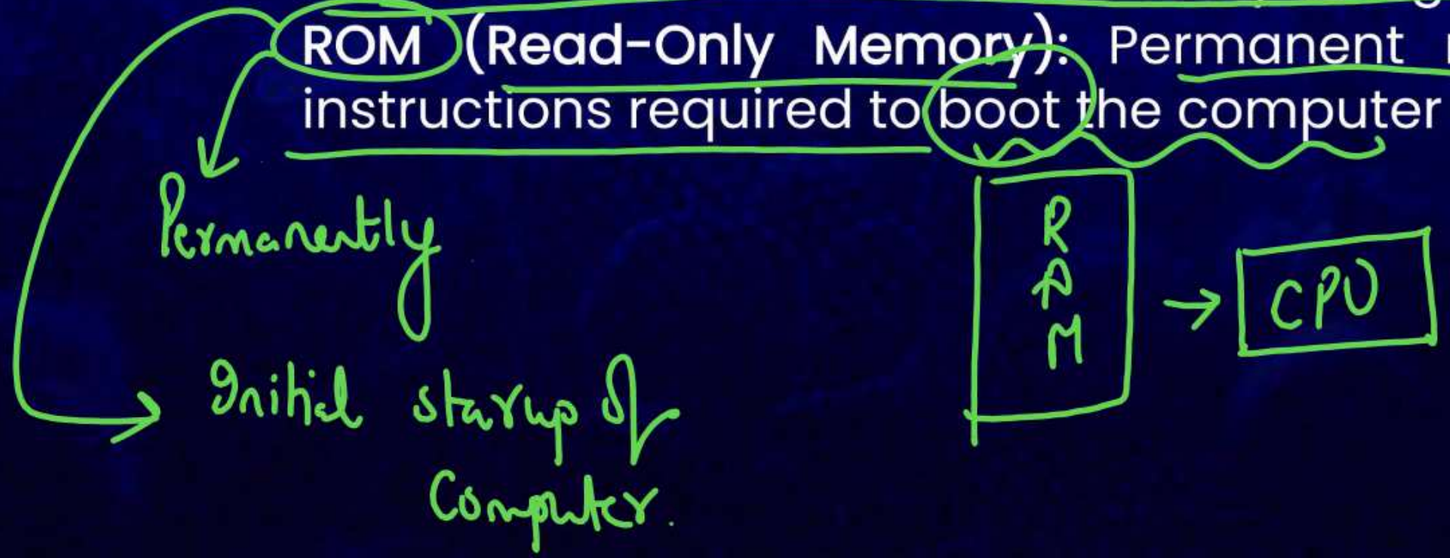
A computer device comprises various elements that facilitate its effective functioning and data processing. Five fundamental components play a pivotal role in simplifying data processing and enhancing convenience. Let's delve into each component along with its functions:

1. **Input Unit:** The input unit allows users to enter data and instructions into the computer system. It includes devices such as keyboards, mice, touchscreens, and scanners. These devices convert physical input (such as keystrokes or mouse movements) into digital signals that the computer can process.

2. **Output Unit:** The output unit displays processed data and results to the user in a human-readable form. Common output devices include monitors, printers, and speakers. They convert digital signals from the computer into understandable output, such as text, images, or sound.
3. **Memory Unit:** The memory unit stores data and instructions temporarily or permanently during the computer's operation. It consists of two main types of memory:

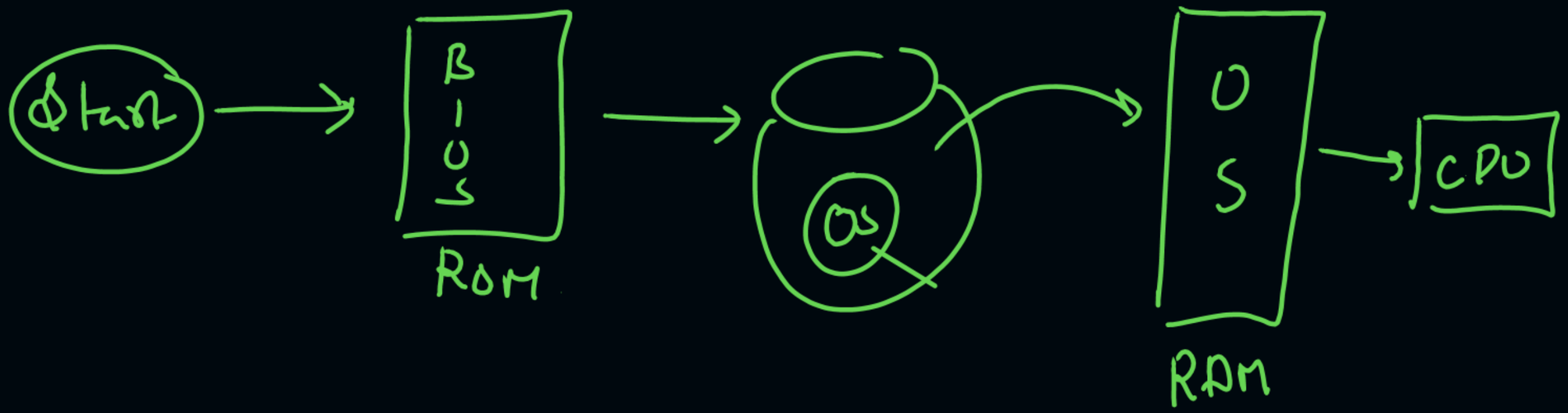
RAM (Random Access Memory): Temporary memory used by the CPU to store data and instructions currently being processed.

ROM (Read-Only Memory): Permanent memory that stores essential instructions required to boot the computer and perform basic functions.



SAMSUNG

Microsoft Windows



4. **Control Unit:** The control unit coordinates and controls the activities of other components within the computer system. It fetches instructions from memory, decodes them, and executes them using the arithmetic and logical unit (ALU). The control unit ensures that data is processed correctly and in the correct sequence.
5. **Arithmetic and Logical Unit (ALU):** The ALU performs arithmetic and logical operations on data as instructed by the control unit. It can perform tasks such as addition, subtraction, multiplication, division, and comparison operations. The ALU is the primary component responsible for executing computational tasks within the computer system.

Calculations

Topic : Basic Computer Fundamentals and Organization

Central Processing Unit (CPU)

- The CPU is the brain of the computer, responsible for executing instructions and processing data. It performs arithmetic, logic, and control operations. Key features include clock speed, number of cores, and cache size.

Motherboard

- The motherboard is the main circuit board of the computer, connecting all components together. It houses the CPU, RAM, storage devices, and other essential components. Key elements include the chipset, expansion slots, and connectors.

Random Access Memory (RAM)

- RAM is temporary memory used by the CPU to store data and instructions currently being processed. It is volatile, meaning it loses its contents when the computer is turned off. RAM size and speed affect system performance.

Topic : Basic Computer Fundamentals and Organization

Storage Devices

- Storage devices store data permanently or temporarily. Common types include:
  - **Hard Disk Drives (HDDs):** Traditional storage devices that use spinning disks to read/write data.
  - **Solid-State Drives (SSDs):** Modern storage devices that use flash memory for faster performance.
  - **Optical Drives:** Used for reading and writing data to optical discs like CDs, DVDs, and Blu-ray discs.

SSD | HDD

Storage capacity ↑

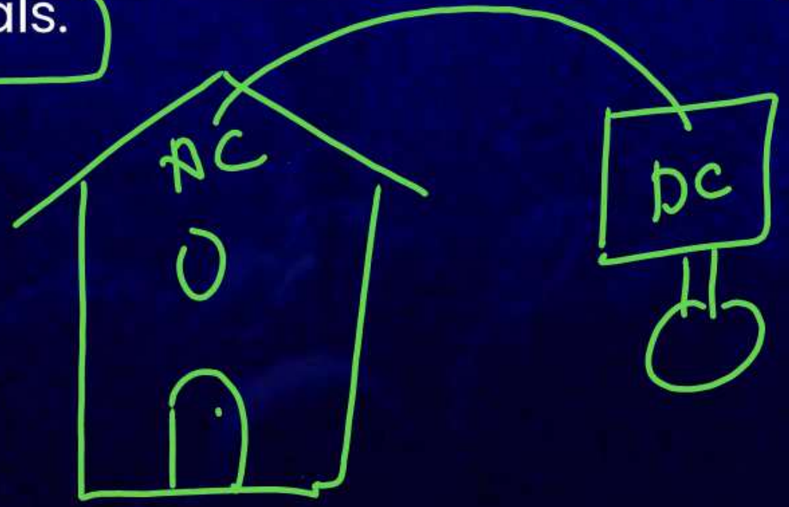
Graphics Processing Unit (GPU)

- The GPU is responsible for rendering images and videos. It accelerates graphics rendering, making it essential for gaming, video editing, and other graphics-intensive tasks. Some CPUs have integrated graphics, while others require a separate GPU.

Power Supply Unit (PSU)

- The PSU converts AC power from the outlet into DC power for the computer's components. It supplies power to the motherboard, CPU, GPU, storage devices, and other peripherals.

Ext. SMPS



Topic : Basic Computer Fundamentals and Organization

Input Devices

- Input devices allow users to interact with the computer. Common examples include:
  - **Keyboard:** Used for typing text and executing commands.
  - **Mouse:** Used for pointing, clicking, and navigating graphical user interfaces.
  - **Touchpad:** Commonly found on laptops, used for cursor movement and gesture control.
  - **Touchscreen:** Allows direct interaction with the display by touching it.

Topic : Basic Computer Fundamentals and Organization

Output Devices

- Output devices display information generated by the computer. Examples include:

Monitor: Displays text, images, and videos generated by the computer.

Printer: Outputs documents and images onto paper. Hardcopy.

Speakers: Output audio generated by the computer.

Input	O/P
Keyboard	
mouse	

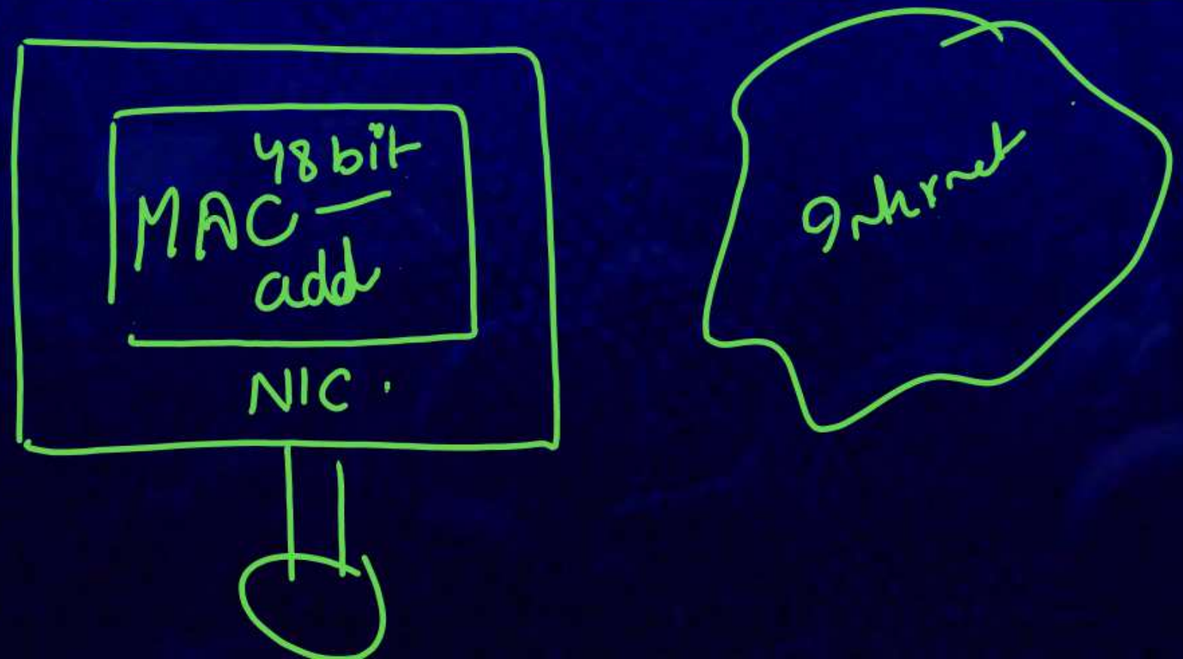
Networking Components

- Networking components allow computers to communicate with each other and access the internet. Key components include:

Network Interface Card (NIC): Allows the computer to connect to a network, either wired or wirelessly.

Router: Routes network traffic between different devices and networks.

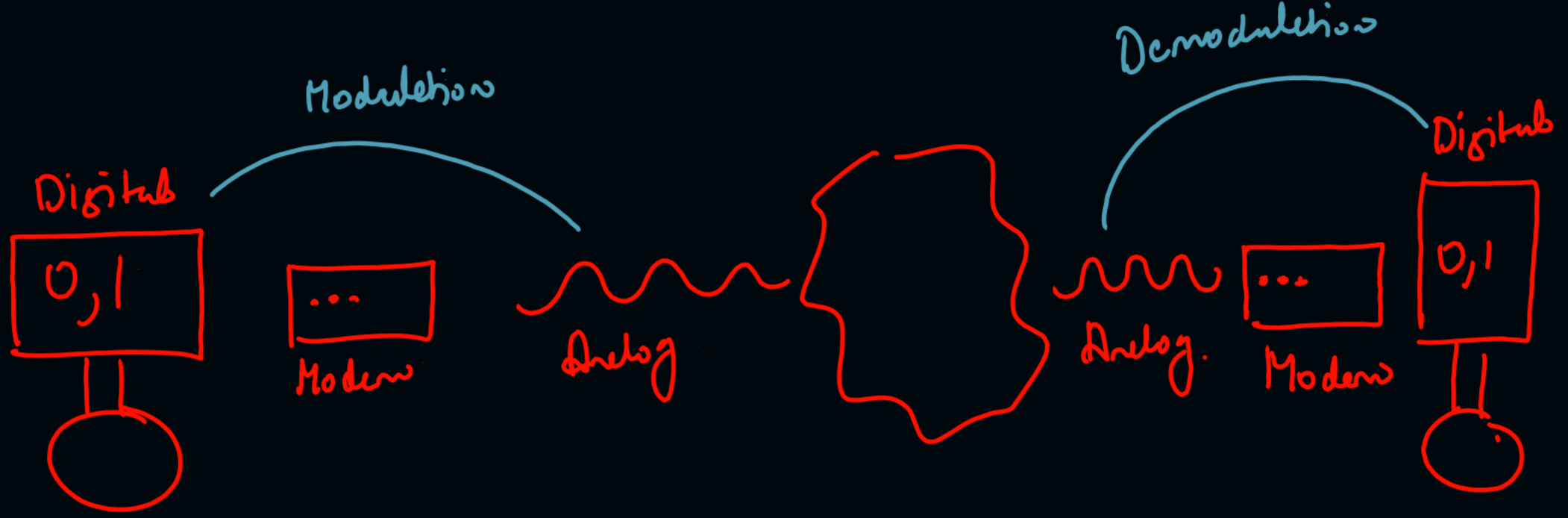
Modem: Converts digital data from the computer into signals that can be transmitted over a network connection.



Wired NIC — Ethernet cable.

Wireless NIC — Radio waves.
Wifi
Bluetooth.


Jio
Company.
ISP
Internet
Service
Provider.
Modem
Router
Phone.
Laptop.
mic
Camera.


Modulation
Demodulation
Digital
Digital
0,1
...
Modem
Analog
Analog
Modem
0,1

..

Cooling System

- Computers generate heat during operation, which can damage components if not properly cooled. Cooling systems include:
  Fans: Circulate air inside the computer case to dissipate heat.
  Heat Sinks: Absorb heat from components like the CPU and GPU, dissipating it into the surrounding air.
  Liquid Cooling Systems: Use liquid coolant to transfer heat away from components, offering more efficient cooling.

Expansion Cards

- Expansion cards add functionality to the computer by providing additional features or connectivity options. Common types include:

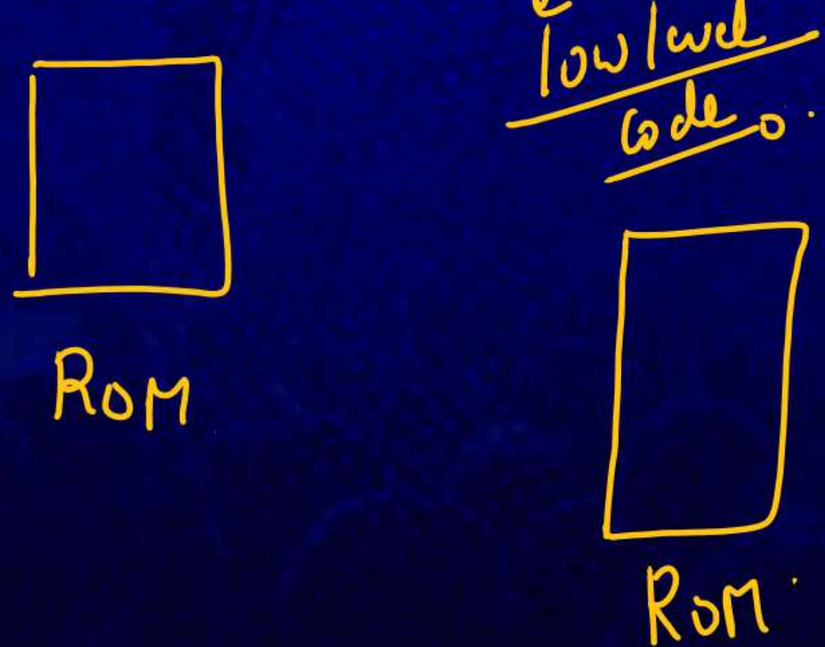
  Graphics Cards: Add dedicated graphics processing power for gaming and multimedia tasks.

  Sound Cards: Enhance audio quality and provide additional audio connectivity options.

  Network Interface Cards: Add wired or wireless networking capabilities to the computer.

BIOS/UEFI

- The Basic Input/Output System (BIOS) or Unified Extensible Firmware Interface (UEFI) is firmware that initializes hardware components during the boot process. It also provides configuration options and controls basic system functions.



Topic : Basic Computer Fundamentals and Organization

Types of Memory in Computer

- In simple terms, computer memory can be categorized into two main types: primary and secondary memory. When focusing on primary memory alone, there are various types available. Some examples of primary memory include the following:

Primary Memory or RAM (Random Access Memory)

- A computer's primary memory, also known as Random Access Memory (RAM), is a fast and volatile type of memory. It holds data and information while our computer is running, but everything stored in RAM is lost when the computer is turned off. RAM plays a crucial role in enhancing the overall performance of the computer by allowing the CPU (Central Processing Unit) to quickly read and write data. This enables efficient multitasking and ensures smooth system performance. The size of RAM, or primary memory, is measured in gigabytes (GB) or terabytes (TB).

Topic : Basic Computer Fundamentals and Organization

- There are two types of RAM: DRAM and SRAM. DRAM stands for Dynamic Random Access Memory, and it's a type of RAM used for dynamic data storage. Each cell in DRAM holds one bit of information and is made up of a tiny capacitor and transistor. However, because of its volatile nature, the information in DRAM needs to be continuously refreshed to prevent data loss. On the other hand, SRAM stands for Static Random Access Memory. It also stores data in memory but of a static nature, meaning the data remains active as long as there is a power supply. Although the same-sized SRAM chip can hold less data compared to DRAM, it has an advantage: it doesn't require continuous refreshing like DRAM does.

ROM (Read Only Memory)

- Read-Only Memory (ROM) is a permanent type of storage in computers. It can only read the information stored in it and cannot be modified or written to. The data remains intact even after the computer is powered off, making it non-volatile. There are five main types of ROM:

M- Maskable

MROM: The oldest type of ROM where the data is set during the manufacturing process, and users cannot change it.

PROM: A digital ROM that allows writing information or programs only once using a special device.

P- Programmable

Flash ROM: This ROM can be programmed in smaller units called sectors or blocks and is commonly used for data transfer between computers and digital devices.

E - Erasable.

- **EPROM:** Data in this ROM can be erased and reprogrammed once. It can store data for around 10-20 years, and to reprogram it, a UV light is used for 40 minutes.
- **EEPROM:** The full form of EEPROM is Electrically Erasable and Programmable Read Only Memory. It allows data to be erased and reprogrammed multiple times using electrical charges.

Electrically

Erasable

Secondary Memory

- Secondary memory is a type of non-volatile storage used for long-term data storage in computers. It includes various storage devices like hard disk drives (HDDs), solid-state drives (SSDs), optical discs (CDs, DVDs, Blu-ray), USB flash drives, and memory cards. These secondary storage devices offer larger storage capacities compared to RAM but have slower access times. Here are the types of secondary memory:

  Hard Disk: A permanent type of computer memory that stores programs, files, and data. It is located on the computer's motherboard and retains data even during power outages or when the system is turned off.

Topic : Basic Computer Fundamentals and Organization

- **Compact Disc (CD):** An optical disk storage device that can store various types of data, including audio, video, files, and other information. CDs use light to read and write data.
- **Pen Drive (USB Flash Drive):** A portable device used as secondary memory in computers for permanently storing data. It is commonly known as a USB flash drive and is used for storing and transferring data.

Cache Memory

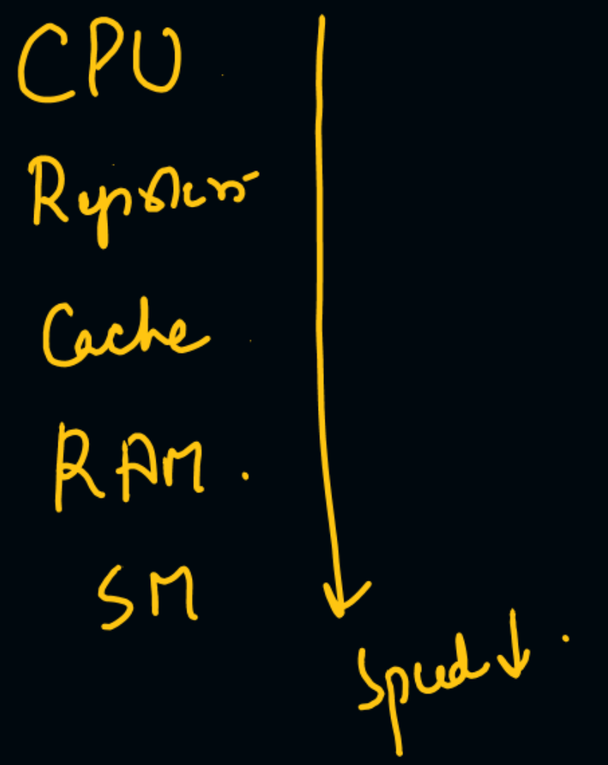
→ Frequently executed data.

- Cache memory in a computer is a small-sized chip-based memory located between the main memory and CPU, very close to the CPU chip. Its purpose is to boost the CPU's performance by providing high-speed temporary storage for frequently accessed data. All the instructions and data that the CPU frequently uses are stored in the Cache memory. There are three types of Cache memory: L1, L2, and L3 cache, each offering different levels of speed and capacity to optimize the computer's overall efficiency.

We will discuss later

Aim → CPU performance ↑

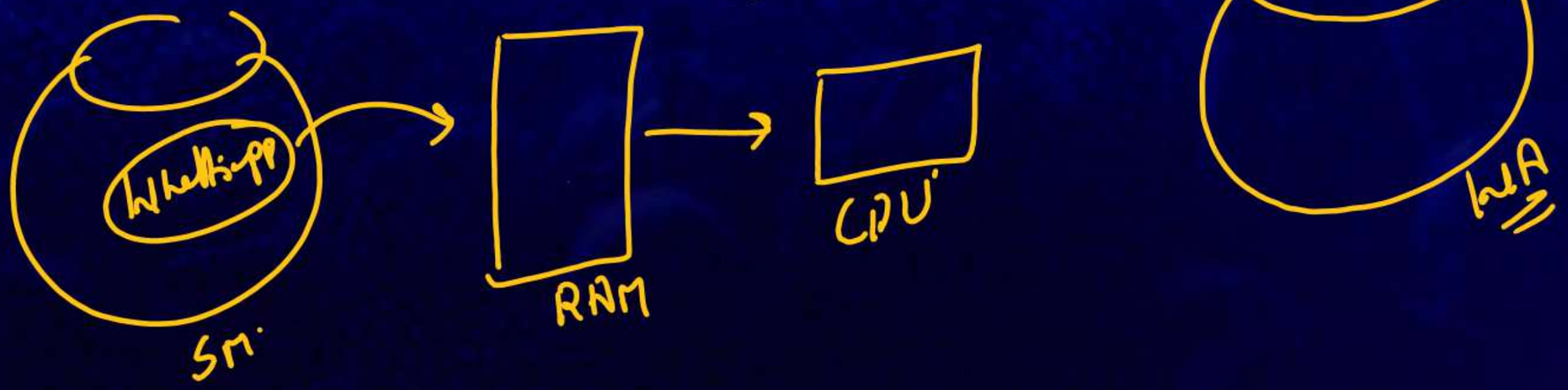



CPU
Registers
Cache
RAM
SM
Speed

Virtual Memory — Not a Physical memory

Is a Concept.

- Virtual memory is like an extension of your computer's RAM. When your computer's RAM is not enough to handle all the programs running at once, the operating system uses a part of the hard drive as virtual memory. This temporary storage helps to hold data that would normally be stored in RAM. Thanks to virtual memory, your computer can handle larger programs and multitask more effectively, ensuring smoother performance even when dealing with numerous tasks simultaneously.



2 Mins Summary

↳ Computer fundamentals

THANK YOU

Ye Tuesday - 11:00 AM